RESEARCH SYNOPSIS / EDITORIAL EDITION 2026

Inhibitory Interactions in Financial Event Streams

Multivariate Hawkes processes, calibration, and parameter identifiability

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Original group report: Modélisation des phénomènes d'inhibition sur les marchés financiers (8 June 2024).
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Research question

Can a multivariate point process with both positive and negative interactions describe and calibrate the timing of different order-book event types? How does the assumed decay rate affect parameter recovery and the geometry of the likelihood?

Mathematical model

For event types $i = 1, \dots, d$, with counting processes N_j, baseline rates $\mu_i > 0$, interaction amplitudes $\alpha_{ij} \in \mathbb{R}$ and decay rates $\beta_{ij} > 0$, the model uses a rectified conditional intensity

$$\lambda_i(t) = \left[\mu_i + \sum_{j=1}^{d} \int_{[0,t)} \alpha_{ij} e^{-\beta_{ij}(t-s)} \, \mathrm{d}N_j(s) \right]_+, \qquad [x]_+ = \max(x, 0). \tag{1}$$

The sign of α_{ij} distinguishes excitation from inhibition. The parameter β_{ij} is a *temporal decay rate*, not the CAPM market beta or the sensitivity of a security's return to a market index. For an observed event history over $[0, T]$, the log-likelihood (up to irrelevant conventions) is

$$\ell(\theta) = \sum_{i=1}^{d} \left[\sum_{k : T_k^{(i)} \le T} \log \lambda_i(T_k^{(i)} -) - \int_0^T \lambda_i(s) \, \mathrm{d}s \right]. \tag{2}$$

The compensator must integrate the *rectified* intensity, accounting for intervals in which inhibition drives the latent linear predictor below zero. For the restricted case $\beta_{ij} = \beta_i$ within each target component, the latent predictor evolves exponentially between successive event times; this permits an analytic zero-crossing calculation.

What the 2024 project examined

The report considered simulation-based parameter estimation, box-plot comparisons of fitted versus generating parameters, order-book event series in several dimensions, and the effect of varying decay assumptions. Its likelihood-profile diagnostics highlighted weak identification of some decay parameters in the tested configurations. These are methodological observations about the studied samples, not claims of a universally consistent or universally inconsistent estimator.

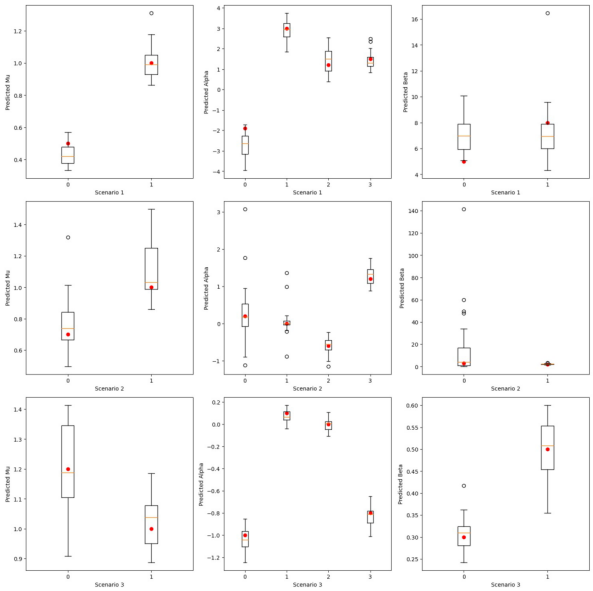


FIGURE 2 – Comparaison avec les véritables paramètres

Figure 1: Parameter-estimation box plots from the 2024 group report (original figure 2). Preserved for historical documentation; the underlying runs have not been independently rerun for this synopsis.

Interpretation and scope

The study concerns *event intensities*, not a directly simulated stochastic differential equation for asset prices. In particular, parameter-decay changes do not by themselves prove a change in asset-price volatility or forecast performance. A limited set of numerical tests may indicate poor conditioning, whereas general convergence claims require stronger assumptions and a controlled sequence of sample sizes. Figures inherited from the report document the historical analysis; no new empirical or numerical results are asserted here.

Research relevance

This project introduced questions that recur in inverse problems: model misspecification, nonlinear likelihood surfaces, distinction between a theoretically identifiable parameter and a numerically stable estimate, and reproducibility of simulation-based calibration.

Selected references

1. A. Bonnet, M. Martinez Herrera, and M. Sangnier, “Inference of multivariate exponential Hawkes processes with inhibition and application to neuronal activity,” *Statistics and Computing* 33 (2023).

2. X. Lu and F. Abergel, “High dimensional Hawkes processes for limit order books: modelling, empirical analysis and numerical calibration,” *Quantitative Finance* 18 (2018).